CLASS 9 · CBSE · GANITA MANJARI PART 2 · SHORT NOTES

Chapter 13: Two Variables, One Line

Topic-wise short notes with solved examples

A linear equation in two variables has infinitely many solutions, and they form a straight line. This chapter covers the slope of a line and the ways of solving a pair of linear equations: substitution, elimination and graphs.

1 Linear equations in two variables

- An equation that can be written as $ax + by + c = 0$, where a, b, c are real numbers and a, b are not both zero, is a **linear equation in two variables**. This is its **standard form**.
- a is the coefficient of x, b is the coefficient of y and c is the constant.
- Fractions can be cleared by multiplying the whole equation by a suitable number.
- An equation in one variable is a special case: $4x + 7 = 0$ is $4x + 0 \cdot y + 7 = 0$.

Example 1.1 Write $3x = 2y - 5$ and $4y = 9$ in standard form and state a, b and c.

SOLUTION

$3x - 2y + 5 = 0$ and $0 \cdot x + 4y - 9 = 0$

$a = 3, b = -2, c = 5$; **and** $a = 0, b = 4, c = -9$

Example 1.2 A fruit seller sells mangoes at ₹80 per kg and bananas at ₹50 per kg. A customer buys x kg of mangoes and y kg of bananas and pays ₹340. Write an equation.

SOLUTION

Cost of mangoes $= 80x$ and cost of bananas $= 50y$

$80x + 50y = 340$

2 Solutions and their graph

- A **solution** is a pair of values (x, y) that satisfies the equation. It is an **ordered pair**: the x-value is written first.
- A linear equation in two variables has **infinitely many** solutions. Choose any value of x and solve for y.
- Plotted as points, all the solutions lie on one **straight line**, and every point of that line is a solution.
- Easy points: put $x = 0$ to find where the line cuts the y-axis, and $y = 0$ to find where it cuts the x-axis.
- If $c = 0$, the line $ax + by = 0$ passes through the origin.
- Multiplying an equation by a non-zero number does not change its solutions: $3x + 4y = 7$ and $6x + 8y = 14$ are the same line.

Example 2.1 Check whether $(3, 2)$ and $(2, 3)$ are solutions of $2x + 3y = 12$. Find the points where its graph cuts the axes.

SOLUTION

$(3, 2)$: $6 + 6 = 12$. $(2, 3)$: $4 + 9 = 13 \neq 12$.

$x = 0$ gives $y = 4$, and $y = 0$ gives $x = 6$.

$(3, 2)$ is a solution and $(2, 3)$ is not. The line cuts the axes at $(0, 4)$ and $(6, 0)$.

Example 2.2 $(2, -1)$ is a solution of both $3ax + 2y = 10$ and $x - by = 5$. Find a and b.

SOLUTION

First equation: $6a - 2 = 10$, so $a = 2$.

Second equation: $2 + b = 5$, so $b = 3$.

$a = 2, b = 3$

3 Slope of a line

- The **slope** (gradient) measures the steepness of a line. Moving from one point of the line to another, the horizontal change is the **run** and the vertical change is the **rise**.

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

- The slope is the same whichever two points of the line are used, and in whichever order.

Slope	The line
Positive	rises from left to right
Negative	falls from left to right
Zero	is horizontal, such as $y = 5$
Undefined	is vertical, such as $x = 2$ (the run is 0)

Example 3.1 Find the slope of the line through (a) $(1, 2)$ and $(5, 10)$ (b) $(-2, 7)$ and $(4, -5)$.

SOLUTION

(a) $\frac{10 - 2}{5 - 1} = \frac{8}{4}$ (b) $\frac{-5 - 7}{4 - (-2)} = \frac{-12}{6}$

(a) 2 (b) -2

Example 3.2 A ramp rises 1.5 m over a horizontal length of 6 m. Find its slope. How long must a ramp of the same slope be to rise 2 m?

SOLUTION

Slope $= \dfrac{1.5}{6} = 0.25$

Run $= \dfrac{2}{0.25}$

Slope 0.25; a horizontal length of 8 m

4 Slope-intercept form

- $y = mx + d$ is the **slope-intercept form**. m is the slope and d is the **y-intercept**: the line cuts the y-axis at $(0, d)$.
- Why m is the slope: for two points on the line, $y_2 - y_1 = (mx_2 + d) - (mx_1 + d) = m(x_2 - x_1)$.
- From the standard form $ax + by + c = 0$ with $b \neq 0$: $y = -\dfrac{a}{b}x - \dfrac{c}{b}$, so

$$m = -\frac{a}{b} \qquad d = -\frac{c}{b}$$

- Two lines with the same slope and different y-intercepts are parallel.

Example 4.1 Find the slope and the y-intercept of $4x + 2y = 10$.

SOLUTION

$2y = -4x + 10$, so $y = -2x + 5$.

Slope -2; y-intercept 5. For each increase of 1 in x, y falls by 2.

5 Pair of linear equations

- Two linear equations in the same two variables form a **pair**: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$.
- A **solution of the pair** is an ordered pair that satisfies **both** equations.
- The idea of every method is to reduce the pair to one equation in one variable.

6 Substitution method

- **Step 1**: From one equation, express one variable in terms of the other.
- **Step 2**: Substitute this expression into the other equation and solve.
- **Step 3**: Put the value back to find the other variable. Check in both equations.

Example 6.1 Solve $2x + 3y = 13$ and $x - y = 4$.

SOLUTION

From the second equation, $x = y + 4$.

Substitute in the first: $2(y + 4) + 3y = 13$, so $5y = 5$ and $y = 1$.

Then $x = 1 + 4 = 5$. Check: $10 + 3 = 13$ and $5 - 1 = 4$.

$x = 5, y = 1$

Example 6.2 Two angles of a triangle are x° and y° with $y = 3x$, and the third angle is 60°. Find x and y.

SOLUTION

$x + y + 60 = 180$. Substituting $y = 3x$: $4x = 120$.

$x = 30, y = 90$

7 Elimination method

- **Step 1**: Multiply the equations by suitable numbers so that one variable has the same coefficient in both.
- **Step 2**: Add or subtract the equations to remove that variable, and solve.
- **Step 3**: Put the value back to find the other variable.

Example 7.1 Solve $3x + 2y = 16$ and $5x - 3y = 14$.

SOLUTION

Multiply the first by 3 and the second by 2: $9x + 6y = 48$ and $10x - 6y = 28$.

Adding: $19x = 76$, so $x = 4$. Then $12 + 2y = 16$ gives $y = 2$.

$x = 4, y = 2$

Example 7.2 3 pens and 2 notebooks cost ₹95. 2 pens and 5 notebooks cost ₹155. Find the cost of each.

SOLUTION

Let a pen cost ₹x and a notebook ₹y: $3x + 2y = 95$ and $2x + 5y = 155$.

Multiply by 2 and by 3: $6x + 4y = 190$ and $6x + 15y = 465$. Subtracting: $11y = 275$, so $y = 25$.

Then $3x = 95 - 50$, so $x = 15$.

A pen costs ₹15 and a notebook costs ₹25.

Example 7.3 The digits of a two-digit number add up to 11. The number formed by reversing the digits is 27 more than the number. Find the number.

SOLUTION

Let the tens digit be x and the units digit be y. The number is $10x + y$ and the reversed number is $10y + x$.

$x + y = 11$ and $(10y + x) - (10x + y) = 27$, that is $y - x = 3$.

Adding: $2y = 14$, so $y = 7$ and $x = 4$.

The number is 47.

Example 7.4 Solve $7x + 5y = 46$ and $5x + 7y = 50$ by adding and subtracting.

SOLUTION

The coefficients are interchanged. Adding: $12x + 12y = 96$, so $x + y = 8$.

Subtracting: $2x - 2y = -4$, so $x - y = -2$.

$x = 3, y = 5$

8 How many solutions?

- During elimination, both variables may disappear together. What is left decides the answer.

Condition	Number of solutions	The two lines
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	exactly one	intersect at a point
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	infinitely many	coincide
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	none	are parallel

- In the second case one equation is a multiple of the other, and elimination gives $0 = 0$. In the third case elimination gives a false statement such as $0 = 1$.

Example 8.1 Without solving, find the number of solutions of (a) $2x + 3y = 7, 4x + 6y = 14$ (b) $2x + 3y = 7, 4x + 6y = 10$ (c) $x + 2y = 5, 3x - y = 1$.

SOLUTION

(a) $\frac{2}{4} = \frac{3}{6} = \frac{7}{14}$ (b) $\frac{2}{4} = \frac{3}{6} \neq \frac{7}{10}$ (c) $\frac{1}{3} \neq \frac{2}{-1}$

(a) infinitely many (b) none (c) exactly one

Example 8.2 For what value of k does the pair $kx + 6y = 9,\ 2x + 3y = 5$ have no solution?

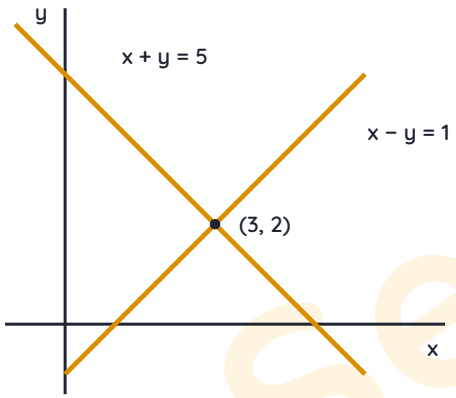
SOLUTION

We need $\frac{k}{2} = \frac{6}{3} \neq \frac{9}{5}$. The first two give $k = 4$, and $2 \neq \frac{9}{5}$.

$k = 4$

9 Graphical method

- Draw the two lines on the same axes. The solutions of the pair are the points that lie on **both** lines.



The lines x + y = 5 and x − y = 1 meet at (3, 2)

- Lines that **intersect** give one solution, the point of intersection. **Parallel** lines give no solution. **Coincident** lines give infinitely many.

Example 9.1 Solve $x + y = 5$ and $x - y = 1$ graphically.

SOLUTION

$x + y = 5$ passes through $(0, 5)$ and $(5, 0)$. $x - y = 1$ passes through $(1, 0)$ and $(0, -1)$.

The two lines cross at $(3, 2)$. Check: $3 + 2 = 5$ and $3 - 2 = 1$.

$x = 3,\ y = 2$

Example 9.2 Plan A costs ₹100 a month plus ₹2 per GB. Plan B costs ₹40 a month plus ₹5 per GB. For what usage do the two plans cost the same?

SOLUTION

For x GB the costs are $y = 100 + 2x$ and $y = 40 + 5x$.

$100 + 2x = 40 + 5x$ gives $x = 20$, and then $y = 140$.

At 20 GB both cost ₹140. Below 20 GB Plan B is cheaper, and above it Plan A is cheaper.

10 Common mistakes

- Writing a solution in the wrong order: $(3, 2)$ means $x = 3$ and $y = 2$.
- Subtracting in different orders in the slope formula. Use $y_2 - y_1$ over $x_2 - x_1$ with the same point first in both.
- Reading the slope of $4x + 2y = 10$ as 4. First write it as $y = -2x + 5$.
- Multiplying only one side of an equation when preparing for elimination.
- Sign errors when subtracting one equation from another. Change the sign of every term.
- Comparing the ratios before writing both equations in the same form.

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